

WELLNESS TOURISM PACKAGE

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Group 4, Using R for Analytics Final Project

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App link: <https://preranadas1407.shinyapps.io/FinalProject/>

GitHub link: https://github.com/wilson51678/R/blob/main/Group%204_Final%20App.R

Video Presentation: <https://purdue.box.com/v/finalrpresentation>

Introduction:

Tours and Travel.com are planning launch a new product for which they required to understand the potential buyers and product preference in their current product catalogue. We conducted Descriptive, Predictive & Prescriptive Analysis to perform the EDA, clustering and optimization to draw the plan for the future and make informed decisions.

Business Problem:

To analyze the customer data clearly to provide insights into the most appropriate customer segmentation and marketing strategies for a new travel package “Wellness Tourism” through a viable business model. The most relevant external stakeholders include customers, and investors while within the organization it is the employees and board members. This problem is amenable to an analytics solution since it has different variables and segments through which we can derive analytical insights and optimize performances. The initial set of tourism business benefits includes bringing direct foreign exchange earnings, contributing heavily to government revenues and offering employment and business opportunities to the local residents. As the data was taken from Kaggle website for this, we believe that either the data was intelligently created to measure the analytical coding ability of coders or is from the stakeholders of “Trips and Travels.com”. We added the constraints that are numeric from the raw data to optimize the value of each variable for the outcome which is product taken by setting the maximum and minimum limitation to variables such as monthly income, age and # of trips etc.

Analytics Problem Definition:

The problem statement in analytical terms is to segment the customers using mathematical models which would help discover groups of similar customers based on finding the smallest variations among customers within each group. In our multivariate linear model, we chose several critical variables such as # of trips, age and the duration of pitch to test the coefficient with the outcome, which is the product taken. The result shows some critical points such as the coefficient of monthly income is negative and the # of trips has a high positive coefficient to the outcome. The key metrics of success are the variables driving product preference, customer segmentation & value optimization. Additionally, we don't have any statement claiming stakeholder agreement on the approach since the dataset is extracted from Kaggle and thus, we cannot say for certain how credible or reliable it is.

Data Description:

Our team discussed the Kaggle data and found it sufficient for our analyses purposes. We also searched for similar datasets in other websites, however nothing concrete was available. Our dataset did not have any prior shiny code. It also didn't have many R codes. So, we took it upon ourselves to analyze and deduce insights on market segmentation. We cleaned the data set by imputing mean for continuous variables such as age, income, and mode for the categorical variables. We further created new variables based on EDA that could be used for clustering analysis. Since the majority of the data was categorical, we didn't have to rescale the data. During our EDA analysis from 3 different directions, we found out important variables that drove customers' choice of product and receptivity to adopt new tourism

products. These variables are - age, income level, marital status, willingness to buy a product based on sales pitch, city tier, occupation and hotel preference.

Methodology Selection:

R provides directly available packages and functions that allow users to explore, model and visualize data and is easy to convert to a client friendly app. The approaches that we selected allow us to learn from past behaviours and understand how they might influence future outcomes by providing actionable insights based on data. For the selection of the graphs, we have mostly used density distribution, bar plots, and pie charts, depending on the variables.

Model Building:

Before the prescriptive analytics, we expected some variables such as monthly income and # of trips to be significant for the outcome of the product taken. As we run the linear model, the result shows that the monthly incomes do not have a positive correlation with outcomes and # of children and person visiting are not significant for our outcome. The highest positive correlation with outcome is # of trips. We then used the result as input parameter in the prescriptive model to figure out the optimization # for each variable. We found that the ideal customer segmentation is 18-year-olds, having ~ 8 trips per year and prefer 40 mins of duration of the pitch.

Functionality:

The ~750 lines code based Shiny App is built on a `dashboardpage()` with foldable sidebar panels, graphs built on `fluidRow`. EDA is performed from 3 angles in 3 tabs - “Customer Data Summary”, “Current User Summary”, and “Potential Buyer Summary”. The “Clustering Analysis” tab performs predictive and prescriptive analysis using user given inputs. While doing this we learned how to add pictures, icons, and beautify the dashboard. We used packages like `shinydashboard`, `shinythemes`, `gbm`, `pROC`, `gbm`, `caret`, `lpSolve`, `lpSolveAPI` to build the dashboard and the analysis. In clustering analysis, we have provided the users the ability to select variables and number of clusters to get linear models and optimize the results. In future we would like to conduct an actual segmentation and segment deep-dive. Further we would like to add more dashboard features and functionalities to make it more dynamic.

References:

Links used for the design part:

- <https://github.com/thepiratex/AttritionAnalysisRShiny/blob/master/App.R>
- https://www.youtube.com/watch?v=6F5_jbBmeJU

Links used for the 5-segment descriptive analysis:

- <https://github.com/BakerSten/Group5FinalR/blob/main/app.R>
- <https://mogali.shinyapps.io/attritionanalysis/>
- <https://github.com/thepiratex/AttritionAnalysisRShiny/blob/master/App.R>